INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study: Android Forensics Investigation of a Google Pixel 3

Lab Title:	**Case 1 Google Pixel 3 Android 10**
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Course Code:	**SBT-DF205**
Course Name:	**Mobile and IoT Forensics Case Study**

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Date: 27/10/2026

Case Brief

Harbour Operations Ltd requests a forensic examination of a company-issued Google Pixel 3 following suspected unauthorised communication and sharing of information. As examiner, you must establish the device users, reconstruct relevant communications, and assess media and location evidence, then explain whether the records support, contradict, or leave the allegation unresolved. Use the original identities and historical dates in the supplied dataset; base every finding on records you have actually examined.

1. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Linux (Kali) or Windows, 40+ GB free	Extract and analyse the dataset.
Archive extraction	unzip / 7z	Expand the ~5.25 GB archive (~9 GB extracted).
Database tools	sqlite3 (CLI) or DB Browser for SQLite	Query AOSP and app databases.
Text editor / grep	nano, grep, less	Read XML config and search for keywords.
Metadata	ExifTool	Extract photo/media metadata and GPS.
Spreadsheet editor	LibreOffice Calc / Excel	Build the Evidence Register and Timeline CSVs.
Hashing	sha256sum	Preserve evidence integrity.

2. Case Folder Setup and Acquire the Evidence

```
┌──(kali㉿kali)-[~/pixel3]
└─$ wget https://digitalcorpora.s3.amazonaws.com/corpora/mobile/andr
oid_10/Non-Cellebrite%20Extraction/Pixel%203.zip
ws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%2
03.zip
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.ama
zonaws.com)|16.15.32.142|:443 ... connected.
HTTP request sent, awaiting response ... 206 Partial Content
Length: 5247820897 (4.9G), 1690737990 (1.6G) remaining [application/
zip]
Saving to: 'Pixel 3.zip.3'

Pixel 3.zip.3    100%[+++++++⟹]   4.89G  1.47MB/s    in 52m 38s

2026-11-01 10:06:02 (523 KB/s) - 'Pixel 3.zip.3' saved [5247820897/5
247820897]
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ ls Pixel\ 3.zip -l
-rw-rw-r-- 1 kali kali 83410816 Oct 31 19:11 'Pixel 3.zip'

┌──(kali㉿kali)-[~/pixel3]
└─$ hashdeep Pixel\ 3.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/kali/pixel3
## $ hashdeep Pixel 3.zip
##
83410816,e1ef1f5090965361b7aac910f4363395,8abc6c57d53fe43dbaf7847c71
4b06b4fad7dbe1e057946325b502a51aed80e1,/home/kali/pixel3/Pixel 3.zip
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ unzip -q "Pixel 3.zip.3" -d pixel3_extracted

┌──(kali㉿kali)-[~/pixel3]
└─$ rm "Pixel 3.zip" "Pixel 3.zip.1" "Pixel 3.zip.2"

┌──(kali㉿kali)-[~/pixel3]
└─$ ls -la pixel3_extracted/
du -sh pixel3_extracted/
total 12
drwxrwxr-x  3 kali kali 4096 Nov  1 18:03  .
drwxrwxr-x  3 kali kali 4096 Nov  1 18:15  ..
drwxrwxr-x 11 kali kali 4096 Feb 15  2020 'Pixel 3'
8.6G    pixel3_extracted/

┌──(kali㉿kali)-[~/pixel3]
└─$ find pixel3_extracted/ -maxdepth 2 -type d
pixel3_extracted/
pixel3_extracted/Pixel 3
pixel3_extracted/Pixel 3/product
pixel3_extracted/Pixel 3/data
pixel3_extracted/Pixel 3/res
pixel3_extracted/Pixel 3/apex
pixel3_extracted/Pixel 3/metadata
pixel3_extracted/Pixel 3/sbin
pixel3_extracted/Pixel 3/mnt
pixel3_extracted/Pixel 3/vendor
pixel3_extracted/Pixel 3/system
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sha256sum 'Pixel 3.zip.3' | tee reports/original_download_sha256.txt
mv 'Pixel 3.zip.3' evidence/Pixel3.zip
chmod 444 evidence/Pixel3.zip
ls -la evidence/Pixel3.zip
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  Pixel 3.zip.3
-r--r--r-- 1 kali kali 5247820897 Nov 21  2020 evidence/Pixel3.zip
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ ls -lah ~/pixel3
total 32K
drwxrwxr-x  8 kali kali 4.0K Nov  2 16:52  .
drwx------ 48 kali kali 4.0K Nov  3 06:39  ..
drwxrwxr-x  2 kali kali 4.0K Nov  2 08:44  evidence
drwxrwxr-x  4 kali kali 4.0K Nov  2 09:12  exported
drwxrwxr-x 11 kali kali 4.0K Feb 15  2020 'Pixel 3'
-rw-r--r--  1 kali kali    0 Nov  2 16:52  pixel3.db
drwxrwxr-x  3 kali kali 4.0K Nov  1 18:03  pixel3_extracted
drwxrwxr-x  2 kali kali 4.0K Nov  2 11:26  reports
drwxrwxr-x  3 kali kali 4.0K Nov  2 10:16  working
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ DB="$HOME/pixel3/exported/mmssms/mmssms.db"

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$DB" ".tables"
addr              attachments          drm   pdu             pending_msgs  raw  sms_restricted  threads  words_content  words_segments
android_metadata  canonical_addresses  part  pdu_restricted  rate          sms  sr_pending      words    words_segdir

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$DB" ".schema"
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE pdu (_id INTEGER PRIMARY KEY AUTOINCREMENT,thread_id INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,msg_box INTEGER,read IN
TEGER DEFAULT 0,m_id TEXT,sub TEXT,sub_cs INTEGER,ct_t TEXT,ct_l TEXT,exp INTEGER,m_cls TEXT,m_type INTEGER,v INTEGER,m_size INTEGER,pri I
NTEGER,rr INTEGER,rpt_a INTEGER,resp_st INTEGER,st INTEGER,tr_id TEXT,retr_st INTEGER,retr_txt TEXT,retr_txt_cs INTEGER,read_status INTEGE
R,ct_cls INTEGER,resp_txt TEXT,d_tm INTEGER,d_rpt INTEGER,locked INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, seen INTEGER DEFAULT 0,creat
or TEXT,text_only INTEGER DEFAULT 0);
CREATE TABLE sqlite_sequence(name,seq);
CREATE TABLE addr (_id INTEGER PRIMARY KEY,msg_id INTEGER,contact_id INTEGER,address TEXT,type INTEGER,charset INTEGER);
CREATE TABLE part (_id INTEGER PRIMARY KEY AUTOINCREMENT,mid INTEGER,seq INTEGER DEFAULT 0,ct TEXT,name TEXT,chset INTEGER,cd TEXT,fn TEXT
,cid TEXT,cl TEXT,ctt_s INTEGER,ctt_t TEXT,_data TEXT,text TEXT);
CREATE TABLE rate (sent_time INTEGER);
CREATE TABLE drm (_id INTEGER PRIMARY KEY,_data TEXT);
CREATE VIEW pdu_restricted AS SELECT * FROM pdu WHERE (msg_box=1 OR msg_box=2) AND (m_type=130)
/* pdu_restricted(_id,thread_id,date,date_sent,msg_box,read,m_id,sub,sub_cs,ct_t,ct_l,exp,m_cls,m_type,v,m_size,pri,rr,rpt_a,resp_st,st,tr
_id,retr_st,retr_txt,retr_txt_cs,read_status,ct_cls,resp_txt,d_tm,d_rpt,locked,sub_id,seen,creator,text_only) */;
CREATE TABLE sms (_id INTEGER PRIMARY KEY,thread_id INTEGER,address TEXT,person INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,protocol
INTEGER,read INTEGER DEFAULT 0,status INTEGER DEFAULT -1,type INTEGER,reply_path_present INTEGER,subject TEXT,body TEXT,service_center TEX
```

3. Task 1 - Evidence Preservation

Original archive details

```
┌──(kali㉿kali)-[~/pixel3]
└─$ stat ~/pixel3/evidence/Pixel3.zip
  File: /home/kali/pixel3/evidence/Pixel3.zip
  Size: 5247820897      Blocks: 10249664   IO Block: 4096   regular file
Device: 8,1     Inode: 3564058     Links: 1
Access: (0444/-r--r--r--)  Uid: ( 1000/    kali)   Gid: ( 1000/    kali)
Access: 2026-11-03 08:51:34.081280141 -0500
Modify: 2020-11-21 22:14:48.000000000 -0500
Change: 2026-11-02 08:44:55.814027423 -0500
 Birth: 2026-11-01 06:45:40.674995408 -0500
```

Archive SHA-256

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sha256sum ~/pixel3/evidence/Pixel3.zip
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  /home/kali/pixel3/evidence/Pixel3.zip

┌──(kali㉿kali)-[~/pixel3]
└─$ cat ~/pixel3/reports/original_archive_sha256.txt
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  /home/kali/pixel3/evidence/Pixel3.zip
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ cd ~/pixel3

echo "═══ HASH BEFORE EXTRACTION ═══"
cat reports/original_download_sha256.txt

echo "═══ HASH AFTER EXTRACTION ═══"
cat reports/archive_sha256_after.txt

echo "═══ COMPARE HASHES ═══"
diff <(awk '{print $1}' reports/original_download_sha256.txt) \
     <(awk '{print $1}' reports/archive_sha256_after.txt) \
&& echo "MATCH: Archive integrity confirmed" \
|| echo "WARNING: Hashes do not match"
═══ HASH BEFORE EXTRACTION ═══
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  Pixel 3.zip.3
═══ HASH AFTER EXTRACTION ═══
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
═══ COMPARE HASHES ═══
MATCH: Archive integrity confirmed
```

Tool versions

```
┌──(kali㉿kali)-[~/pixel3]
└─$ unzip -v | head -3
sqlite3 --version
exiftool -ver
sha256sum --version | head -1
UnZip 6.00 of 20 April 2009, by Debian. Original by Info-ZIP.

Latest sources and executables are at ftp://ftp.info-zip.org/pub/infozip/ ;
3.53.4 2026-07-24 19:02:57 bf7c7f30031888f4e796e429ab3978879485813aaca6f641c7b33e4e0945alt1 (64-bit)
13.55
sha256sum (GNU coreutils) 9.10
```

Evidence register for important databases

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f \( \
-iname "mmssms.db*" -o \
-iname "contacts2.db*" -o \
-iname "calllog.db*" -o \
-iname "bugle_db*" -o \
-iname "msgstore.db*" -o \
-iname "wa.db*" \
\) -print0 | sort -z | while IFS= read -r -d '' file; do
size=$(stat -c %s "$file")
hash=$(sha256sum "$file" | awk '{print $1}')
printf '"%s",%s,%s\n' "$file" "$size" "$hash"
done > ~/pixel3/reports/database_evidence_register.csv
```

Field	Finding
Case reference	SBT-DF205-C1
Source	Digital Corpora – Android 10
Archive name	Pixel3.zip
Original downloaded name	Pixel 3.zip.3
Archive size	5,247,820,897 bytes
Examiner	Yusuf Oluwaseun Dada
Unzip	6.00
SQLite	3.53.4
ExifTool	13.55
SHA-256	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
Original hash record	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a reports/original_download_sha256.txt
After-extraction hash record	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a reports/archive_sha256_after.txt

4. Task 2 - Device Identity and Profiles

```
┌──(kali㉿kali)-[~/pixel3]
└─$ SYS="$HOME/pixel3/pixel3_extracted/Pixel 3/system/build.prop"
PROD="$HOME/pixel3/pixel3_extracted/Pixel 3/product/build.prop"
VEND="$HOME/pixel3/pixel3_extracted/Pixel 3/vendor/build.prop"

grep -nE 'ro.build.id=|ro.build.version.release=|ro.build.version.security_patch=' "$SYS"

grep -nE 'ro.product.product.(manufacturer|model|device)=|ro.product.build.fingerprint=' "$PROD"

grep -nE 'ro.product.vendor.(manufacturer|model|device)=|ro.vendor.build.fingerprint=' "$VEND"
21:ro.build.id=QQ1A.200105.003
29:ro.build.version.release=10
30:ro.build.version.security_patch=2020-01-01
6:ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
14:ro.product.product.device=blueline
15:ro.product.product.manufacturer=Google
16:ro.product.product.model=Pixel 3
17:ro.vendor.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
25:ro.product.vendor.device=blueline
26:ro.product.vendor.manufacturer=Google
27:ro.product.vendor.model=Pixel 3
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ BASE="$HOME/pixel3/pixel3_extracted/Pixel 3/data/system/users"

cat "$BASE/userlist.xml"
cat "$BASE/0.xml"
cat "$BASE/11.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
    <user id="11" />
</users>
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1
A.200105.003/6079943:user/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
    <name>This Is</name>
    <restrictions />
</user>
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/b
lueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ for DB in ~/pixel3/working/accounts/*.db; do
echo "==== $DB ===="
sqlite3 "$DB" ".schema accounts"
done
==== /home/kali/pixel3/working/accounts/accounts_ce.db ====
CREATE TABLE accounts ( _id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT NOT NULL, type TEXT NOT NULL, password TEXT, UNIQUE(name,type));
CREATE TRIGGER accountsDelete DELETE ON accounts BEGIN   DELETE FROM authtokens     WHERE accounts_id=OLD._id ;   DELETE FROM extras     W
HERE accounts_id=OLD._id ; END;
==== /home/kali/pixel3/working/accounts/accounts_de.db ====
CREATE TABLE accounts ( _id INTEGER PRIMARY KEY, name TEXT NOT NULL, type TEXT NOT NULL, previous_name TEXT, last_password_entry_time_mill
is_epoch INTEGER DEFAULT 0, UNIQUE(name,type));
CREATE TRIGGER accountsDelete DELETE ON accounts BEGIN   DELETE FROM grants     WHERE accounts_id=OLD._id ; END;
CREATE TRIGGER accountsDeleteVisibility DELETE ON accounts BEGIN   DELETE FROM visibility     WHERE accounts_id=OLD._id ; END;
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ for DB in ~/pixel3/working/accounts/*.db; do
echo "==== $DB ===="
sqlite3 -header -column "$DB" "SELECT name,type FROM accounts;"
done
==== /home/kali/pixel3/working/accounts/accounts_ce.db ====
name                  type
--------------------  ------------------------------------
+19195794674          com.viber.voip
858233690             org.telegram.messenger
Duo                   com.google.android.apps.tachyon
Messenger             com.facebook.messenger
Signal                org.thoughtcrime.securesms
Skype                 com.skype.raider
TDfir                 com.twitter.android.auth.login
TextNow               com.enflick.android.TextNow.account
TikTok                com.zhiliaoapp.musically
WhatsApp              com.whatsapp
imo HD                com.imo.android.imous
thisisdfir            com.silentcircle.account
thisisdfir@gmail.com  com.google
==== /home/kali/pixel3/working/accounts/accounts_de.db ====
name                  type
--------------------  ----------
thisisdfirtwo@gmail.com  com.google
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ cat "$BASE/0/registered_services/android.accounts.AccountAuthenticator.xml"

cat "$BASE/11/registered_services/android.accounts.AccountAuthenticator.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<services>
    <service uid="10239" type="com.enflick.android.TextNow.account" />
    <service uid="10244" type="com.whatsapp" />
    <service uid="10187" type="com.google.android.gm.pop3" />
    <service uid="10141" type="com.google" />
    <service uid="10209" type="com.google.android.apps.tachyon" />
    <service uid="10238" type="com.zhiliaoapp.account" />
    <service uid="10245" type="com.twitter.android.auth.login" />
    <service uid="10224" type="com.silentcircle.account" />
    <service uid="10236" type="com.skype.raider" />
    <service uid="10241" type="com.facebook.messenger" />
    <service uid="10187" type="com.google.android.gm.exchange" />
    <service uid="10238" type="com.zhiliaoapp.musically" />
    <service uid="10231" type="com.viber.voip" />
    <service uid="10141" type="com.google.android.gms.matchstick" />
    <service uid="10240" type="com.imo.android.imous" />
    <service uid="10233" type="org.thoughtcrime.securesms" />
    <service uid="10234" type="org.telegram.messenger" />
    <service uid="10187" type="com.google.android.gm.legacyimap" />
</services>
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<services>
    <service uid="1110187" type="com.google.android.gm.pop3" />
    <service uid="1110141" type="com.google" />
    <service uid="1110141" type="com.google.android.gms.matchstick" />
```

Item	Finding
Manufacturer / Model	Google Pixel 3. Supported by build fingerprint google/blueline/blueline. The direct manufacturer/model properties were not returned.
Android version	Android 10
Build ID	QQ1A.200105.003
Security patch	2020-01-01
User profiles found	Two profiles: User ID 0 (This Is) and User ID 11 (User 2).
Account corroborated and its second source	thisisdfir@gmail.com appears in Chrome preferences and Google Play Store preferences.
Limits of attributing activity to an individual	A user profile or email account identifies an OS-level or app-level account. It does not conclusively prove who operated the device at a given time, especially if the device was shared.

5. Task 3 - Settings and Connectivity

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted/Pixel\ 3/data/system \
-type f \( -iname "*settings*.xml" -o -iname "*settings*.db" \) \
-printf '%s bytes  %p\n'
5593 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml
21631 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_secure.xml
33305 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_config.xml
25315 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_global.xml
5374 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_ssaid.xml
4864 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/11/settings_system.xml
16542 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/11/settings_secure.xml
1611 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/11/settings_ssaid.xml
20480 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/locksettings.db
238 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/watchlist_settings.xml
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ grep -RinE 'screen_brightness|screen_off_timeout|location_mode|airplane_mode_on|adb_enabled|bluetooth' \
~/pixel3/pixel3_extracted/Pixel\ 3/data/system/users/0/ \
2>/dev/null
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/package-restrictions.xml:597:    <pkg name="com.android.bluetooth.auto_gene
rated_rro_vendor__" ceDataInode="2899" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/package-restrictions.xml:632:            <item name="io.presage.core.receiv
er.BluetoothReceiver" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/package-restrictions.xml:1607:            <item name="com.android.settings.
Settings$BluetoothSettingsActivity" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/package-restrictions.xml:1715:    <pkg name="com.android.bluetoothmidiservi
ce" ceDataInode="3206" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/package-restrictions.xml:1756:    <pkg name="com.android.bluetooth" ceDataI
node="3243" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml:15:  <setting id="16" name="screen_brightness_mode" val
ue="1" package="android" defaultValue="1" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml:19:  <setting id="15" name="screen_brightness_for_vr" v
alue="111" package="android" defaultValue="111" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml:25:  <setting id="6" name="volume_bluetooth_sco" value=
"7" package="android" defaultValue="7" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml:30:  <setting id="55" name="screen_off_timeout" value="
300000" package="com.android.settings" defaultValue="300000" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_system.xml:46:  <setting id="877" name="screen_brightness" value="
101" package="android" defaultValue="101" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_secure.xml:13:  <setting id="146" name="location_mode" value="3" p
ackage="com.google.android.gms" defaultValue="3" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_secure.xml:77:  <setting id="65" name="bluetooth_name" value="Pixe
l 3" package="android" defaultValue="Pixel 3" defaultSysSet="true" />
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/users/0/settings_secure.xml:122:  <setting id="71" name="bluetooth_address" value="
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted/Pixel\ 3/data \
-type f \( -iname "*wifi*" -o -iname "*network*" \) \
-printf '%s bytes  %p\n'
2 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.thinkyeah.galleryvault/files/adc3/data/network_requests
65 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.photos/shared_prefs/gms_icing_mdd_network_usage_mon
itor.xml
175 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/Wifistats.xml
176 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/com.google.android.gms.checkin.interna
l.network_backoff_shared_pref_name.xml
121 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/WifistatsPH.xml
65 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/CAST_ACTIVE_NETWORK_MAP.xml
65 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/gms_icing_mdd_network_usage_monitor.xml
11130 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/Network Persistent State
40960 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/Network Action Predictor
16384 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/jp.naver.line.android/databases/network_usage_log
8192 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.zhiliaoapp.musically/files/keva/repo/smart_network_select/smart_netwo
rk_select.blk
4096 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.zhiliaoapp.musically/files/keva/repo/smart_network_select/smart_netwo
rk_select.chk
12332 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.zhiliaoapp.musically/files/offline/365b20e8f6c343df1eff65214a0e3e74/
tiktokm_fission/resource/task_page_i18n/images/error_network_huoshan.2a5c9c23.png
88 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.facebook.orca/files/http/historical/orca_network_map
65 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.googlequicksearchbox/shared_prefs/gms_icing_mdd_network_
usage_monitor.xml
20 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.ims/cache/_asset_artifacts_wifi_call_quality_inference_m
odel_artifacts_dnn_v1_inference_state.ckp1647436529668069729.sha1
20 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.ims/cache/_asset_artifacts_wifi_call_quality_inference_m
odel_artifacts_dnn_v1_inference_state.ckp2000151791564851807.sha1
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted/Pixel\ 3/data \
-type f -iname "*bluetooth*" \
-printf '%s bytes  %p\n'
544 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/resource-cache/vendor@overlay@Bluetooth__auto_generated_rro_vendor.apk@idmap
166 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.gms/shared_prefs/bluetooth_addresses_prefs.xml
16456 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/dalvik-cache/arm64/system@app@BluetoothMidiService@BluetoothMidiService.apk@c
lasses.vdex
3247800 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/dalvik-cache/arm64/system@app@Bluetooth@Bluetooth.apk@classes.vdex
8192 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/dalvik-cache/arm64/system@app@BluetoothMidiService@BluetoothMidiService.apk@cl
asses.art
147456 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/dalvik-cache/arm64/system@app@Bluetooth@Bluetooth.apk@classes.art
118 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.bluetooth/shared_prefs/bluetooth_volume_map.xml
398 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.bluetooth/shared_prefs/com.android.bluetooth_preferences.
xml
210152 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db-wal
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db-shm
4096 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db
335 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/slice/client_android.permission.BLUETOOTH_ADMIN@0
22104 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/package_cache/3fd1af0b70135ef9f203f6fe5902df9bd5769a96/Bluetooth-16
2568 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/package_cache/3fd1af0b70135ef9f203f6fe5902df9bd5769a96/BluetoothMidiSer
vice-16
1592 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/system/package_cache/3fd1af0b70135ef9f203f6fe5902df9bd5769a96/Bluetooth__auto_
generated_rro_vendor.apk-16
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ grep -RilE 'SSID|BSSID|PreSharedKey|DeviceName|Address' \
~/pixel3/pixel3_extracted/Pixel\ 3/data/misc/wifi \
~/pixel3/pixel3_extracted/Pixel\ 3/data/misc/bluetooth \
2>/dev/null
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/misc/wifi/WifiConfigStore.xml
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ echo "═══ BLUETOOTH BACKUP ═══"
sed -n '1,240p' "$BASE/data/misc/bluedroid/bt_config.bak"
═══ BLUETOOTH BACKUP ═══
[Info]
FileSource = Empty
TimeCreated = 2020-01-29 15:38:12

[Metrics]
Salt256Bit = 8c68fa3461653b701e23f6bac4ae6fa71e56f7610758288c0fc27eb
7228379ba

[Adapter]
Address = 7c:d9:5c:ac:a2:ce
LE_LOCAL_KEY_IRK = 195d5c0827a16e7e95b8780cc4243049
LE_LOCAL_KEY_IR = 6aec03cd0258857111ffe969e845ec72
LE_LOCAL_KEY_DHK = e506ba23332577b905649764bcd9e248
LE_LOCAL_KEY_ER = f6fdc1e629635e0d40e479d02694a0ab
ScanMode = 0
DiscoveryTimeout = 120
```

Item	Findings
Setting 1: Wi-Fi	wifi_on = 1 – Wi-Fi was recorded as enabled.
Setting 2: Bluetooth	bluetooth_on = 0 – Bluetooth was recorded as disabled.
Additional setting: Location	location_mode = 3; providers: gps,network.
Additional setting: USB debugging	adb_enabled = 1 – USB debugging was enabled in the recorded settings.
Wi-Fi SSID found	CcookiesDcastleR5 – appears in the MacAddressMap entry with security type WPA_PSK.
Wi-Fi MAC/BSSID entry	ae:f6:c7:a8:a5:c2
Bluetooth device identifier	Rouge – address b4:ec:02:73:ff:93.
Phone Bluetooth identifier	Pixel 3 – address 7c:d9:5c:ac:a2:ce.
Saved config vs. proven connection	The files contain stored Wi-Fi and Bluetooth records. They do not prove a successful connection at a particular time.

6. Task 4 - AOSP Communications and Contacts

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f -iname "mmssms.db*" \
-printf '%s bytes  %p\n'
118784 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db
118784 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.providers.telephony/databases/mmssms.db
118784 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.android.providers.telephony/databases/mmssms.db
118784 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/user_de/0/com.android.providers.telephony/databases/mmss
ms.db
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f -iname "mmssms.db" \
-exec sha256sum {} \;
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.provide
rs.telephony/databases/mmssms.db
d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/user_de/0/com.android.pr
oviders.telephony/databases/mmssms.db
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data
/com.android.providers.telephony/databases/mmssms.db
d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/user
_de/0/com.android.providers.telephony/databases/mmssms.db
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ DB="$HOME/pixel3/exported/mmssms/mmssms.db"

sqlite3 -header -column "$DB" "
SELECT COUNT(*) AS total_sms FROM sms;
SELECT type,COUNT(*) AS total FROM sms GROUP BY type;
SELECT COUNT(*) AS total_mms FROM pdu;
"
total_sms
──────────
24
type  total
────  ─────
1     22
2     2
total_mms
──────────
7
```

```
└─$ sqlite3 -header -column "$DB" "
SELECT
_id,
thread_id,
address,
date AS raw_timestamp_ms,
datetime(date/1000,'unixepoch') AS utc_time,
type,
read,
body
FROM sms
ORDER BY date;
"
_id thread_id   address     raw_timestamp_ms      utc_time       type read                              body
  2         4 +13233205525    1580322045251 2020-01-29 18:20:45    1    0 <#> Please enter 304868 into LINE within the next 30 mins.
                                                                          JFoQLtyexga

  3         5 732873          1580322646631 2020-01-29 18:30:46    1    0 Telegram code: 59389

                                                                          You can also tap on this link to log in:
                                                                          https://t.me/login/59389

  4         6 +18437514985    1580322696889 2020-01-29 18:31:36    1    0 <#> Your Signal verification code: 303-077

                                                                          doDiFGKPO1r

  5         7 +12253690855    1580322728551 2020-01-29 18:32:08    1    0 Your imo code is 685614
                                                                          9K+9OFePKE9
```

```
16  16 +13233208934  1580519052798 2020-02-01 01:04:12  1  1 Your Wire code is 172488.

                                                          Open https://account.wire.com/v/172488 to verify your number.
17  13 32665         1580757526393 2020-02-03 19:18:46  1  0 👤 Xavier Mccollum is a new Facebook friend suggestion:
                                                          https://fb.com/l/9OSKBTdMAcfo8C
18  13 32665         1580919741561 2020-02-05 16:22:21  1  0 🔔 ThisIs, you have 1 new notification on Facebook:
                                                          https://fb.com/l/a5ANAeaZdoseLW
19  13 32665         1581063359840 2020-02-07 08:15:59  1  0 🔔 ThisIs, you have 1 new notification on Facebook:
                                                          https://fb.com/l/1KFllApNpuN8igJ
20  13 32665         1581185821854 2020-02-08 18:17:01  1  0 🔔 ThisIs, you have 1 new notification on Facebook:
                                                          https://fb.com/l/aym2DdgkRIfkix
21  17 9195790479    1581273702392 2020-02-09 18:41:42  2  1 I am on my way to the coffee shop. Do you want anything?
22  13 32665         1581444160908 2020-02-11 18:02:40  1  0 🔔 ThisIs, you have 6 new notifications on Facebook:
                                                          https://fb.com/l/1IOHQ5bDdc60Nl9
23  13 32665         1581534738811 2020-02-12 19:12:18  1  0 🔔 ThisIs, you have 9 new notifications on Facebook:
                                                          https://fb.com/l/akGyU7KWPrdRmS
24  13 32665         1581628344576 2020-02-13 21:12:24  1  0 🔔 ThisIs, you have 9 new notifications on Facebook:
                                                          https://fb.com/l/1KypwvpROvYfv7n
25  17 919-579-0479  1581685743277 2020-02-14 13:09:03  2  1 Wow, the messages sync over between accounts.
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$DB" "
SELECT
thread_id,
COUNT(*) AS message_count,
MIN(datetime(date/1000,'unixepoch')) AS first_message_utc,
MAX(datetime(date/1000,'unixepoch')) AS last_message_utc
FROM sms
GROUP BY thread_id
ORDER BY thread_id;
"
```

thread_id	message_count	first_message_utc	last_message_utc
4	1	2020-01-29 18:20:45	2020-01-29 18:20:45
5	3	2020-01-29 18:30:46	2020-01-29 20:16:10
6	1	2020-01-29 18:31:36	2020-01-29 18:31:36
7	1	2020-01-29 18:32:08	2020-01-29 18:32:08
8	1	2020-01-29 18:33:40	2020-01-29 18:33:40
9	1	2020-01-29 18:35:03	2020-01-29 18:35:03
10	1	2020-01-29 18:37:08	2020-01-29 18:37:08
11	1	2020-01-29 18:43:08	2020-01-29 18:43:08
12	1	2020-01-29 18:50:13	2020-01-29 18:50:13
13	8	2020-01-29 18:50:24	2020-02-13 21:12:24
14	1	2020-01-29 21:15:33	2020-01-29 21:15:33
15	1	2020-02-01 01:00:03	2020-02-01 01:00:03
16	1	2020-02-01 01:04:12	2020-02-01 01:04:12
17	2	2020-02-09 18:41:42	2020-02-14 13:09:03

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$DB" "
SELECT
_id,
thread_id,
address,
body,
date AS raw_timestamp_ms,
datetime(date/1000,'unixepoch') AS utc_time,
type,
read
FROM sms
WHERE thread_id=17
ORDER BY date;
"
```

_id	thread_id	address	body	raw_timestamp_ms	utc_time	type	read
21	17	9195790479	I am on my way to the coffee shop. Do you want anything?	1581273702392	2020-02-09 18:41:42	2	1
25	17	919-579-0479	Wow, the messages sync over between accounts.	1581685743277	2020-02-14 13:09:03	2	1

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$DB" "
SELECT
_id,
thread_id,
date AS raw_timestamp,
datetime(date,'unixepoch') AS utc_time,
m_type,
msg_box,
sub
FROM pdu
ORDER BY date;
"
```

_id	thread_id	raw_timestamp	utc_time	m_type	msg_box	sub
1	17	1580587493	2020-02-01 20:04:53	128	2	
2	17	1580587564	2020-02-01 20:06:04	132	1	
3	17	1580587672	2020-02-01 20:07:52	128	2	
4	17	1580587692	2020-02-01 20:08:12	132	1	
5	17	1580587767	2020-02-01 20:09:27	128	2	
6	17	1581273788	2020-02-09 18:43:08	132	1	
7	17	1581685808	2020-02-14 13:10:08	132	1	

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$DB" "
SELECT
_id,
mid,
seq,
ct,
name,
text,
_data
FROM part
ORDER BY mid;
"
```

_id	mid	seq	ct	name	text	_data
1	1	-1	application/smil		<smil><head><layout><root-layout/><region id="Text" top="0" left="0" height="100%" width="100%"/></layout></head><body><par dur="5000ms"><text src="text000001.txt" region="Text" /></par></body></smil>	
2	1	0	text/plain		Here's Signal.	
3	2	0	text/plain	body	It looks like the messaging is run through the Messages app. Strange.	
4	3	-1	application/smil		<smil><head><layout><root-layout/><region id="Text" top="0" left="0" height="100%" width="100%"/></layout></head><body><par dur="5000ms"><text src="text000001.txt" region="Text" /></par></body></smil>	
5	3	0	text/plain		It might be an attempt at unified messaging ... ?	
6	4	0	text/plain	body	Maybe?	
7	5	-1	application/smil		<smil><head><layout><root-layout/><region id="Text" top="0" left="0" height="100%" width="100%"/></layout></head><body><par dur="5000ms"><text src="text000001.txt" region="Text" /></par></body></smil>	
8	5	0	text/plain		It will be interesting to see where the messages are stored.	
9	6	0	text/plain	body	No, thank you	
10	7	0	text/plain	body	That is interesting.	

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f \( \
-iname "contacts2.db" -o -iname "calllog.db" \
\) -printf '%s bytes  %p\n'
376832 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db
40960 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db
376832 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.android.providers.contacts/databases/contacts2.db
40960 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.android.providers.contacts/databases/calllog.db
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ CDB="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db"

sqlite3 "$CDB" ".tables"
_sync_state            deleted_contacts      photo_files            search_index_stat    view_groups               view_v1_organizations
_sync_state_metadata   directories           pre_authorized_uris    settings             view_metadata_sync        view_v1_people
accounts               groups                presence               status_updates       view_metadata_sync_state  view_v1_phones
agg_exceptions         metadata_sync         properties             stream_item_photos   view_raw_contacts         view_v1_photos
agg_presence           metadata_sync_state   raw_contacts           stream_items         view_raw_entities         visible_contacts
android_metadata       mimetypes             search_index           v1_settings          view_stream_items
contacts               name_lookup           search_index_content   view_contacts        view_v1_contact_methods
data                   nickname_lookup       search_index_docsize   view_data            view_v1_extensions
data_usage_stat        packages              search_index_segdir    view_data_usage      view_v1_group_membership
default_directory      phone_lookup          search_index_segments  view_entities        view_v1_groups
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$CDB" "
SELECT _id,display_name FROM raw_contacts;
"
_id  display_name
---  ------------
1    Josh Hickman
2    Josh Hickman
3    Josh Hickman
4    Josh Hickman
5    Josh Hickman
6    Josh Hickman
7    Josh Hickman
8    Josh Hickman
9    Josh Hickman
10   Josh Hickman
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f -iname "calllog.db" -print
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db
/home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.android.providers.contacts/databases/calllog.db
```

database structure

```
┌──(kali㉿kali)-[~/pixel3]
└─$ CDB="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db"

sqlite3 "$CDB" ".tables"
android_metadata     calls     properties     voicemail_status
```

Call records count

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$CDB" ".schema"
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE properties (property_key TEXT PRIMARY KEY, property_value TEXT );
CREATE TABLE calls (_id INTEGER PRIMARY KEY AUTOINCREMENT,number TEXT,presentation INTEGER NOT NULL DEFAULT 1,post_dial_digits TEXT NOT NU
LL DEFAULT '',via_number TEXT NOT NULL DEFAULT '',date INTEGER,duration INTEGER,data_usage INTEGER,type INTEGER,features INTEGER NOT NULL
DEFAULT 0,subscription_component_name TEXT,subscription_id TEXT,phone_account_address TEXT,phone_account_hidden INTEGER NOT NULL DEFAULT 0
,sub_id INTEGER DEFAULT -1,new INTEGER,name TEXT,numbertype INTEGER,numberlabel TEXT,countryiso TEXT,voicemail_uri TEXT,is_read INTEGER,ge
ocoded_location TEXT,lookup_uri TEXT,matched_number TEXT,normalized_number TEXT,photo_id INTEGER NOT NULL DEFAULT 0,photo_uri TEXT,formatt
ed_number TEXT,add_for_all_users INTEGER NOT NULL DEFAULT 1,last_modified INTEGER DEFAULT 0,call_screening_component_name TEXT,call_screen
ing_app_name TEXT,block_reason INTEGER NOT NULL DEFAULT 0,_data TEXT,has_content INTEGER,mime_type TEXT,source_data TEXT,source_package TE
XT,transcription TEXT,transcription_state INTEGER NOT NULL DEFAULT 0,state INTEGER,dirty INTEGER NOT NULL DEFAULT 0,deleted INTEGER NOT NU
LL DEFAULT 0,backed_up INTEGER NOT NULL DEFAULT 0,restored INTEGER NOT NULL DEFAULT 0,archived INTEGER NOT NULL DEFAULT 0,is_omtp_voicemai
l INTEGER NOT NULL DEFAULT 0);
CREATE TABLE sqlite_sequence(name,seq);
CREATE TABLE voicemail_status (_id INTEGER PRIMARY KEY AUTOINCREMENT,source_package TEXT NOT NULL,phone_account_component_name TEXT,phone_
account_id TEXT,settings_uri TEXT,voicemail_access_uri TEXT,configuration_state INTEGER,data_channel_state INTEGER,notification_channel_st
ate INTEGER,quota_occupied INTEGER DEFAULT -1,quota_total INTEGER DEFAULT -1,source_type TEXT);

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$CDB" "
SELECT COUNT(*) AS total_calls FROM calls;
"
total_calls
-----------
14
```

Call records extraction

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$CDB" "
SELECT
_id,
number,
type,
date AS raw_timestamp_ms,
datetime(date/1000,'unixepoch') AS utc_time,
duration,
new,
name
FROM calls
ORDER BY date;
"
_id  number        type  raw_timestamp_ms  utc_time             duration  new  name
---  ------------  ----  ----------------  -------------------  --------  ---  ----------
1                  3     1580483796007     2020-01-31 15:16:36  0         0
2                  3     1580494070836     2020-01-31 18:07:50  0         0
3                  3     1580502115252     2020-01-31 20:21:55  0         0
4                  3     1580585280414     2020-02-01 19:28:00  0         0
5                  3     1580758342425     2020-02-03 19:32:22  0         0
6   +332629243087  3     1581192896305     2020-02-08 20:14:56  0         0
7   +19195121037   3     1581521849939     2020-02-12 15:37:29  0         0
8   +19195121037   4     1581521901000     2020-02-12 15:38:21  3         0
9   +15755863247   3     1581555164000     2020-02-13 00:52:44  0         0
10  +15755863247   4     1581555213000     2020-02-13 00:53:33  3         0
11  9195790479     2     1581643741051     2020-02-14 01:29:01  11        1    Josh Hickman
12  +19195790479   1     1581643824061     2020-02-14 01:30:24  65        1    Josh Hickman
13  9195790479     2     1581685557104     2020-02-14 13:05:57  86        1    Josh Hickman
14  9195790479     1     1581691278351     2020-02-14 14:41:18  91        1    Josh Hickman
```

Item	Findings
mmssms.db copies compared (paths, hashes, row counts)	Four copies were identified. The main database and Magisk mirror copy had matching SHA-256 hashes (cd5452f...53f1342) and 24 SMS rows each. The two user_de copies also had matching hashes (d0ce903...ef34c827) and zero SMS rows each.
5+ SMS records / one reconstructed	Thread ID 13 contained eight SMS records, including Facebook confirmation and notification messages dated 29

Item	Findings
conversation (thread_id)	January to 13 February 2020. Thread ID 17 also contained two sent SMS messages dated 9 and 14 February 2020.
One reconstructed MMS (pdu + part)	Seven PDU records and ten linked part records were found. PDU records were linked to their parts using pt.mid = p._id. Thread ID 17 contained reconstructed message content, including messages about Signal and unified messaging.
2+ numbers linked to contacts	The contact Josh Hickman was linked to number formats 9195790479 and +1 919-579-0479. Other stored formats included (919) 579-0479.
3 calls examined	Three calls involving the stored Josh Hickman number were examined on 14 February 2020: one outgoing call lasting 11 seconds, one incoming call lasting 65 seconds, and another outgoing call lasting 86 seconds.
Calendar Provider tables and row counts	The database calendar.db contained 14 tables of interest. Calendars had 1 row, Colors had 35, and CalendarCache had 4. The Events, EventsRawTimes, Instances, Attendees, Reminders, CalendarAlerts and ExtendedProperties tables had zero rows.

7. Task 5 - Google Messages and Dialer

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f -iname "bugle_db*" \
-printf '%s bytes  %p\n'
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/bugle_db-shm
465592 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/bugle_db-wal
425984 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/bugle_db
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.google.android.apps.messaging/databases/bugle_db
-shm
465592 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.google.android.apps.messaging/databases/bugle_d
b-wal
425984 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.google.android.apps.messaging/databases/bugle_d
b
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ SRC="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases"

mkdir -p ~/pixel3/working/bugle

cp "$SRC/bugle_db" "$SRC/bugle_db-wal" "$SRC/bugle_db-shm" \
~/pixel3/working/bugle/

BDB="$HOME/pixel3/working/bugle/bugle_db"

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$BDB" ".tables"
add_contact_banner              etouffee_rcs_metadata           p2p_conversation_suggestion_event  reports_concat_view
cms                             events                          participants                       self_participants
cms_notifications               flagged_messages                parts                              user_references
conversation_frecency_view      generic_worker_queue            pending_incoming_message_rcs_table  user_refs_concat_view
conversation_image_parts_view   link_preview                    rbm_payment_request_metadata       verified_sms_blacklisted_senders
conversation_participants       link_preview_participants_table read_reports                       verified_sms_brands
conversation_suggestions        linked_account                  received_message_phone_numbers     verified_sms_senders
conversations                   message_spam                    recent_expressive_stickers         work_queue
desktop                         message_status                  recent_gifs
disambiguations                 messages                        reminders
draft_parts_view                messages_annotations            remote_instances
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$BDB" ".schema"
CREATE TABLE conversations (_id INTEGER PRIMARY KEY AUTOINCREMENT, sms_thread_id INT DEFAULT(0), name TEXT, name_is_automatic BOOLEAN DEFA
ULT(1), latest_message_id INT, snippet_text TEXT, subject_text TEXT, preview_uri TEXT, preview_content_type TEXT, show_draft INT DEFAULT(0
), draft_snippet_text TEXT, draft_subject_text TEXT, draft_preview_uri TEXT, draft_preview_content_type TEXT, etouffee_default INT DEFAULT
(0), archive_status INT DEFAULT(0), sort_timestamp INT DEFAULT(0), last_read_timestamp INT DEFAULT(0), icon TEXT, participant_contact_id I
NT DEFAULT(-1), participant_lookup_key TEXT, participant_normalized_destination TEXT, current_self_id TEXT, participant_count INT DEFAULT(
0), notification_enabled INT DEFAULT(1), notification_sound_uri TEXT, notification_vibration INT DEFAULT(1), include_email_addr INT DEFAUL
T(0), sms_service_center TEXT, participant_id_list TEXT, source_type INT DEFAULT(0), rcs_session_id INT DEFAULT(-1), join_state INT DEFAUL
T(0), conv_type INT DEFAULT(0), send_mode INT DEFAULT(0), IS_ENTERPRISE INT DEFAULT(0), has_ea2p_bot_recipient INT DEFAULT(0), last_intera
ctive_event_timestamp INT DEFAULT(0), participant_display_destination TEXT, spam_warning_dismiss_status INT DEFAULT(0), open_count INT DEF
AULT(0), last_logged_scooby_metadata_timestamp INT DEFAULT(0), delete_timestamp INT DEFAULT(0), cms_id TEXT, rcs_group_id TEXT, rcs_confer
ence_uri TEXT);
CREATE TABLE sqlite_sequence(name,seq);
CREATE INDEX index_conversations_sms_thread_id ON conversations(sms_thread_id);
CREATE INDEX index_conversations_archive_status ON conversations(archive_status);
CREATE INDEX index_conversations_sort_timestamp ON conversations(sort_timestamp);
CREATE INDEX index_conversations_rcs_session_id ON conversations(rcs_session_id);
CREATE INDEX index_conversations_delete_timestamp ON conversations(delete_timestamp);
CREATE UNIQUE INDEX index_conversations_cms_id ON conversations(cms_id);
CREATE TABLE messages (_id INTEGER PRIMARY KEY AUTOINCREMENT, conversation_id INT, sender_id INT, sent_timestamp INT DEFAULT(0), queue_ins
ert_timestamp INT DEFAULT(0), received_timestamp INT DEFAULT(0), message_protocol INT DEFAULT(0), message_status INT DEFAULT(0), message_r
eport_status INT DEFAULT(0), seen INT DEFAULT(0), read INT DEFAULT(0), sms_message_uri TEXT, sms_priority INT DEFAULT(0), sms_message_size
 INT DEFAULT(0), mms_subject TEXT, mms_transaction_id TEXT, mms_content_location TEXT, mms_expiry INT DEFAULT(0), mms_retrieve_text TEXT,
raw_status INT DEFAULT(0), self_id INT, retry_start_timestamp INT DEFAULT(0), cloud_sync_id TEXT, rcs_message_id STRING, etouffee_status I
NT DEFAULT(0), verification_status INT DEFAULT(0), is_hidden INT DEFAULT(0), rcs_remote_instance STRING, rcs_file_transfer_session_id INT
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$BUGLE" "PRAGMA table_info(messages);"
sqlite3 "$BUGLE" "PRAGMA table_info(parts);"
```

cid	name	type	notnull	dflt_value	pk
0	_id	INTEGER	0	NULL	1
1	conversation_id	INT	0	NULL	0
2	sender_id	INT	0	NULL	0
3	sent_timestamp	INT	0	'0'	0
4	queue_insert_timestamp	INT	0	'0'	0
5	received_timestamp	INT	0	'0'	0
6	message_protocol	INT	0	'0'	0
7	message_status	INT	0	'0'	0
8	message_report_status	INT	0	'0'	0

```
┌──(kali㉿kali)-[~/pixel3]
└─$ ANNOTATED="$ROOT/com.google.android.dialer/databases/annotated_call_log.db"

sqlite3 -header -column "$ANNOTATED" "
SELECT name
FROM sqlite_master
WHERE type='table'
ORDER BY name;
"
      name
-----------------
AnnotatedCallLog
android_metadata

┌──(kali㉿kali)-[~/pixel3]
└─$ for TABLE in $(sqlite3 "$ANNOTATED" "
SELECT name FROM sqlite_master
WHERE type='table' AND name NOT LIKE 'sqlite_%';
"); do
    echo "========== $TABLE =========="
    sqlite3 -header -column "$ANNOTATED" "PRAGMA table_info(\"$TABLE\");"
done
========== android_metadata ==========
cid   name    type  notnull  dflt_value  pk
```

```
└─$ sqlite3 -header -column "$BDB" "
SELECT name FROM sqlite_master
WHERE type='table'
ORDER BY name;
"
                name
------------------------------------
add_contact_banner
cms
cms_notifications
conversation_participants
conversation_suggestions
conversations
desktop
disambiguations
etouffee_rcs_metadata
events
flagged_messages
generic_worker_queue
link_preview
link_preview_participants_table
linked_account
message_spam
message_status
messages
messages_annotations
p2p_conversation_suggestion_event
participants
parts
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$BUGLE" "
SELECT 'messages' AS table_name, COUNT(*) AS row_count FROM messages

UNION ALL
SELECT 'parts', COUNT(*) FROM parts;
"
table_name  row_count
──────────  ─────────
messages    36
parts       31
```

```
sqlite3 -header -column "$CALLDB" "
SELECT
    _id,
    number,
    name,
    datetime(date/1000,'unixepoch') AS call_time_utc,
    duration,
    type
FROM calls
ORDER BY date;
"
_id  number        name         call_time_utc     duration  type
───  ────────────  ───────────  ────────────────  ────────  ────
1                               2020-01-31 15:16:36  0         3
2                               2020-01-31 18:07:50  0         3
3                               2020-01-31 20:21:55  0         3
4                               2020-02-01 19:28:00  0         3
5                               2020-02-03 19:32:22  0         3
6    +332629243087              2020-02-08 20:14:56  0         3
7    +19195121037               2020-02-12 15:37:29  0         3
8    +19195121037               2020-02-12 15:38:21  3         4
9    +15755863247               2020-02-13 00:52:44  0         3
10   +15755863247               2020-02-13 00:53:33  3         4
11   9195790479    Josh Hickman 2020-02-14 01:29:01  11        2
12   +19195790479  Josh Hickman 2020-02-14 01:30:24  65        1
13   9195790479    Josh Hickman 2020-02-14 13:05:57  86        2
14   9195790479    Josh Hickman 2020-02-14 14:41:18  91        1
```

Item	Findings
Google Messages database location and schema tables	com.google.android.apps.messaging/databases/bugle_db. It contains 36 tables, including messages, parts, conversations, and participants.
3+ Messages SMS matched to Telephony Provider records	22 records returned by the timestamp comparison. Two personal SMS messages were also confirmed by matching timestamps and message content.

Item	Findings
Google Dialer database location	com.google.android.dialer/databases/dialer.db.
2+ Dialer calls compared to AOSP call log, differences noted	The separate annotated_call_log.db contains 12 call records. All 12 matched AOSP records by timestamp, duration, and call type. AOSP contains 14 records; the two extra entries are voicemail records.
One result reproduced via SQL	SQL queries successfully reproduced the Google Messages/SMS matches and the Google Dialer/AOSP call comparison.

8. Task 6 - WhatsApp and a Second Application

WhatsApp databases

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f \( \
-iname "msgstore.db*" -o \
-iname "wa.db*" \
\) -printf '%s bytes  %p\n'
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/wa.db-shm
325512 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/wa.db-wal
4096 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/wa.db
823296 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/msgstore.db
449112 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/msgstore.db-wal
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases/msgstore.db-shm
53228 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/WhatsApp/Databases/msgstore.db.crypt12
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/wa.db-shm
325512 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/wa.db-wal
4096 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/wa.db
823296 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/msgstore.db
449112 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/msgstore.db-wal
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/data/com.whatsapp/databases/msgstore.db-shm
53228 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/WhatsApp/Databases/msgstore.db.crypt12
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ SRC="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.whatsapp/databases"

mkdir -p ~/pixel3/working/whatsapp

cp "$SRC/msgstore.db"* ~/pixel3/working/whatsapp/
cp "$SRC/wa.db"* ~/pixel3/working/whatsapp/

WDB="$HOME/pixel3/working/whatsapp/msgstore.db"
WADB="$HOME/pixel3/working/whatsapp/wa.db"
```

```
└─$ sqlite3 "$WDB" ".tables"
available_message_view             message_ftsv2_segdir                          message_template
away_messages                      message_ftsv2_segments                        message_template_button
away_messages_exemptions           message_ftsv2_stat                            message_template_quoted
call_log                           message_future                                message_text
call_log_participant_v2            message_group_invite                          message_thumbnail
chat                               message_link                                  message_thumbnails
chat_list                          message_location                              message_vcard
chat_view                          message_media                                 message_vcard_jid
conversion_tuples                  message_media_interactive_annotation          message_view
deleted_chat_job                   message_media_interactive_annotation_vertex   messages
deleted_messages_ids_view          message_mentions                              messages_dehydrated_hsm
deleted_messages_view              message_orphaned_edit                         messages_fts
frequent                           message_payment                               messages_fts_content
frequents                          message_payment_status_update                 messages_fts_segdir
group_participant_device           message_payment_transaction_reminder          messages_fts_segments
group_participant_user             message_product                               messages_hydrated_four_row_template
group_participants                 message_quoted                                messages_links
group_participants_history         message_quoted_group_invite                   messages_quotes
jid                                message_quoted_group_invite_legacy            messages_vcards
keywords                           message_quoted_location                       messages_vcards_jids
labeled_jid                        message_quoted_media                          missed_call_log_participant
labeled_jids                       message_quoted_mentions                       missed_call_logs
labeled_messages                   message_quoted_product                        pay_transaction
labeled_messages_fts               message_quoted_text                           pay_transactions
labeled_messages_fts_content       message_quoted_vcard                          props
labeled_messages_fts_segdir        message_revoked                               quick_replies
labeled_messages_fts_segments      message_send_count                            quick_reply_attachments
labels                             message_streaming_sidecar                     quick_reply_keywords
legacy_available_messages_view     message_system                                quick_reply_usage
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$WDB" ".schema" > ~/pixel3/reports/whatsapp_schema.txt
cat ~/pixel3/reports/whatsapp_schema.txt
CREATE TABLE messages (_id INTEGER PRIMARY KEY AUTOINCREMENT, key_remote_jid TEXT NOT NULL, key_from_me INTEGER, key_id TEXT NOT NULL, sta
tus INTEGER, needs_push INTEGER, data TEXT, timestamp INTEGER, media_url TEXT, media_mime_type TEXT, media_wa_type TEXT, media_size INTEGE
R, media_name TEXT, media_caption TEXT, media_hash TEXT, media_duration INTEGER, origin INTEGER, latitude REAL, longitude REAL, thumb_imag
e TEXT, remote_resource TEXT, received_timestamp INTEGER, send_timestamp INTEGER, receipt_server_timestamp INTEGER, receipt_device_timesta
mp INTEGER, read_device_timestamp INTEGER, played_device_timestamp INTEGER, raw_data BLOB, recipient_count INTEGER, participant_hash TEXT,
 starred INTEGER, quoted_row_id INTEGER, mentioned_jids TEXT, multicast_id TEXT, edit_version INTEGER, media_enc_hash TEXT, payment_transa
ction_id TEXT, forwarded INTEGER, preview_type INTEGER, send_count INTEGER);
CREATE TABLE sqlite_sequence(name,seq);
CREATE UNIQUE INDEX messages_key_index on messages (key_remote_jid, key_from_me, key_id);
CREATE INDEX messages_jid_id_index on messages (key_remote_jid, _id);
CREATE INDEX media_hash_index on messages (media_hash);
CREATE INDEX media_type_index on messages (media_wa_type);
CREATE INDEX media_type_jid_index on messages (key_remote_jid, media_wa_type);
CREATE INDEX starred_index on messages (starred);
CREATE TABLE message (    _id                     INTEGER PRIMARY KEY AUTOINCREMENT,    chat_row_id             INTEGER NOT NULL,    fro
m_me                INTEGER NOT NULL,    key_id                  TEXT    NOT NULL,    sender_jid_row_id       INTEGER,    status
          INTEGER,    broadcast               INTEGER,    recipient_count         INTEGER,    participant_hash        TEXT,    origi
nation_flags        INTEGER,    origin                  INTEGER,    timestamp               INTEGER,    received_timestamp      INTEGER
,    receipt_server_timestamp INTEGER,    message_type            INTEGER,    text_data               TEXT,    starred
INTEGER,    lookup_tables           INTEGER);
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$WADB" ".tables"
android_metadata               wa_biz_profiles_hours        wa_contact_storage_usage     wa_group_descriptions
system_contacts_version_table  wa_biz_profiles_websites     wa_contacts                  wa_props
wa_biz_profiles                wa_block_list                wa_group_add_black_list      wa_vnames
wa_biz_profiles_categories     wa_contact_capabilities      wa_group_admin_settings      wa_vnames_localized
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 "$WADB" ".schema"
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE wa_contacts (_id INTEGER PRIMARY KEY AUTOINCREMENT, jid TEXT NOT NULL, is_whatsapp_user BOOLEAN NOT NULL, status TEXT, status
_timestamp INTEGER, number TEXT, raw_contact_id INTEGER, display_name TEXT, phone_type INTEGER, phone_label TEXT, unseen_msg_count INTEGER
, photo_ts INTEGER, thumb_ts INTEGER, photo_id_timestamp INTEGER, given_name TEXT, family_name TEXT, wa_name TEXT, sort_name TEXT, nicknam
e TEXT, company TEXT, title TEXT, status_autodownload_disabled INTEGER, keep_timestamp INTEGER, is_spam_reported INTEGER, is_sidelist_sync
ed BOOLEAN DEFAULT 0, is_business_synced BOOLEAN DEFAULT 0);
CREATE TABLE sqlite_sequence(name,seq);
CREATE INDEX is_wa_index ON wa_contacts(is_whatsapp_user);
CREATE INDEX jid_index ON wa_contacts(jid);
CREATE TABLE wa_contact_capabilities (_id INTEGER PRIMARY KEY AUTOINCREMENT, jid TEXT NOT NULL, capability TEXT NOT NULL, value TEXT NOT N
ULL, updated_at INTEGER NOT NULL);
CREATE UNIQUE INDEX wa_contact_capabilities_jid_capability_index ON wa_contact_capabilities(jid,capability);
CREATE TABLE system_contacts_version_table (id INTEGER PRIMARY KEY, version INTEGER);
CREATE TABLE wa_vnames (_id INTEGER PRIMARY KEY AUTOINCREMENT, jid TEXT NOT NULL, serial INTEGER NOT NULL, issuer TEXT NOT NULL, expires I
NTEGER, verified_name TEXT NOT NULL, industry TEXT, city TEXT, country TEXT, verified_level INTEGER, identity_unconfirmed_since INTEGER, c
ert_blob BLOB);
CREATE UNIQUE INDEX vname_jid_index ON wa_vnames(jid);
CREATE TABLE wa_vnames_localized (_id INTEGER PRIMARY KEY AUTOINCREMENT, jid TEXT NOT NULL, lg TEXT NOT NULL, lc TEXT NOT NULL, verified_n
ame TEXT NOT NULL);
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ KIK="$ROOT/kik.android/databases"

┌──(kali㉿kali)-[~/pixel3]
└─$ for DB in "$KIK"/*kikDatabase.db "$KIK"/*kikCoreDatabase.db "$KI
K"/*userRosterEntries.db "$KIK"/*convoProfileEntries.db "$KIK"/*cont
actProfileEntries.db
do
    echo "========== $(basename "$DB") =========="
    sqlite3 "$DB" "
    SELECT name
    FROM sqlite_master
    WHERE type='table'
    ORDER BY name;
    "
done
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -header -column "$WADB" "
SELECT
_id,
jid,
number,
display_name,
wa_name,
is_whatsapp_user
FROM wa_contacts
ORDER BY _id;
"
_id      jid                number      display_name   wa_name        is_whatsapp_u
                                                                     ser
---  ---------------  ----------  -------------  -------------  ---------------
  1  status@                                                                   1
     broadcast

  2  16505434800@s                                                             1
     .whatsapp.net

  3  19195790479@s  9195790479  Josh Hickman   Josh Hickman                    1
     .whatsapp.net
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ for DB in "$KIK"/*userRosterEntries.db "$KIK"/*convoProfileEntri
es.db "$KIK"/*contactProfileEntries.db "$KIK"/*kikCoreDatabase.db
do
    echo "========== $(basename "$DB") =========="
    sqlite3 -header -column "$DB" "
    SELECT * FROM $(sqlite3 "$DB" "SELECT name FROM sqlite_master WH
ERE type='table' AND name NOT LIKE 'android_metadata';" | head -n 1)
;
    "
done
========== d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.userRosterEntries.db
 ==========
          bare_jid                  proto_bytes
----------------------------  ----------------------------

thisisdfir_r3b@talk.kik.com
                              ^p
                              ^Nthisisdfir_r3b^Z^L

                              ThisIsDFIR"
```

Item	Findings
WhatsApp account identified (source file)	The file registration.RegisterPhone.xml contains the phone number 9195794674 and formatted number (919) 579-4674. Verification state is recorded as 0.
Two or more WhatsApp messages	Three readable messages were recovered from msgstore.db involving JID 19195790479@s.whatsapp.net. They include two outgoing messages and one incoming message dated 8 February 2020.
Second app chosen	Kik Messenger.
Account association found	kikCoreDatabase.db identifies the active Kik username as ThisIsDFIR. The associated JID is thisisdfir_r3b@talk.kik.com.
Two or more communication records extracted	Six records were recovered from messagesTable, dated 1 February 2020. Four contain readable message text, while two have blank bodies.
One participant correlated with another source	Josh Hickman appears in Kik contacts, WhatsApp contacts, Android contacts and call records. The matching name supports a cross-source association.
Message-status schema validated for this app version	The messagesTable schema contains an integer field named read_state. All six recovered records contain value 500. The meaning of 500 has not yet been confirmed from the application version or its status definitions.

9. Task 7 - Maps and Photographs

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find ~/pixel3/pixel3_extracted -type f \
-path "*com.google.android.apps.maps*" \
-printf '%s bytes  %p\n'
98526 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/e3758d09c9d6815481b4be5b3e3266b0ee550e5bca8fa01590ceb15caa20c499.0
616 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/249c31fe6a7ff47e62a37213ac65caccca81163dc0b6313b70b86e2583acb848.0
427 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/2a05312bd2db8bf392f820609e27c9e4d1e5cbda6086b7d45bb163062a2fcc78.0
1688 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/journal
340 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/ae870cc783e9c03badd2f7245cd01521ae3b4e2aa0b96f3e4734be545136aa99.0
748 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/a02e79e763ba01ac463e92032b767f2415b6df6c543182b1da2dad63b307ab19.0
117320 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/47b742b3697431173421b8b5948ed9fe5a13ba7ab9e0067912bff97e57bc5258.0
432 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/image_manager_disk_cache/43927d574c236d97439dc9c61fb43b1558d749aa61ca47b5fb3f9c0d40874ccc.0
4 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/cronet-cache-40719148/version
48 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/cronet-cache-40719148/disk_cache/index-dir/the-real-index
24 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/cronet-cache-40719148/disk_cache/index
1620 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/cache/cronet-cache-40719148/prefs/local_prefs.json
3783 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/shared_prefs/settings_preference.xml
131 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/shared_prefs/client_parameters_details.xml
309 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/shared_prefs/ue3Preference.xml
65 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/shared_prefs/cache_settings_preference.xml
130 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/shared_prefs/_has_set_default_values.xml
3231744 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/cache/diskcache/map_cache.db
32768 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/cache/diskcache/map_cache.db-shm
4572 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._TURN_LEFT.mp3
7992 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._DESTINATION_ON_LEFT.mp3
3708 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._TURN_RIGHT.mp3
5976 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._IN_600_FEET.mp3
6012 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._IN_1000_FEET.mp3
5256 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._IN_A_QUARTER_MILE.mp3
5652 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._HEAD_SOUTHEAST.mp3
10908 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/._ROUNDABOUT_TAKE_2ND_EXIT.mp3
593308 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/media/0/Android/data/com.google.android.apps.maps/testdata/voice/en_US2eb0f91b/en_US/hol/voice_instructions_unitless.zip
129212 bytes  /home/kali/pixel3/pixel3_extracted/Pixel 3/sbin/.magisk/mirror/data/system/package_cache/3fd1af0b70135ef9f203f6fe5902df9bd5769a96/com.google.android.apps.maps-L0I1_Q7J82×8vVbOXb6Q7A==-0
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ BASE="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/databases"

for DB in gmm_myplaces.db gmm_storage.db ugc_photos_location_data.db gmm_sync.db ue3.db inbox_notifications.db
do
    echo "========== $DB =========="
    sqlite3 "$BASE/$DB" ".tables"
done
========== gmm_myplaces.db ==========
android_metadata     sync_corpus     sync_item
========== gmm_storage.db ==========
android_metadata     gmm_storage_table
========== ugc_photos_location_data.db ==========
android_metadata     photos_image_labels     photos_top_feature
========== gmm_sync.db ==========
android_metadata     sync_corpus_metadata     sync_item_data     sync_metadata
========== ue3.db ==========
android_metadata     userevent3_table
========== inbox_notifications.db ==========
android_metadata     inboxNotifications
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ mkdir -p ~/pixel3/reports/maps
for DB in gmm_myplaces.db gmm_storage.db ugc_photos_location_data.db gmm_sync.db ue3.db inbox_notifications.dbdo
    sqlite3 "$BASE/$DB" ".schema" > ~/pixel3/reports/maps/"${DB}_schema.txt"done
```

```
└─$ BASE="$HOME/pixel3/pixel3_extracted/Pixel 3/data/data/com.google.android.apps.maps/databases"

for DB in gmm_myplaces.db gmm_storage.db ugc_photos_location_data.db gmm_sync.db ue3.db inbox_notifications.db
do
    echo "========== $DB =========="
    sqlite3 "$BASE/$DB" "SELECT name FROM sqlite_master WHERE type='table';"
done
========== gmm_myplaces.db ==========
┌──────────────────┐
│       name       │
├──────────────────┤
│ android_metadata │
│ sync_item        │
│ sync_corpus      │
└──────────────────┘
========== gmm_storage.db ==========
┌──────────────────┐
│       name       │
├──────────────────┤
│ android_metadata │
│ gmm_storage_table│
└──────────────────┘
========== ugc_photos_location_data.db ==========
┌──────────────────┐
│       name       │
├──────────────────┤
│ android_metadata │
│ photos_image_labels │
│ photos_top_feature │
└──────────────────┘
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ for DB in gmm_myplaces.db gmm_storage.db ugc_photos_location_data.db gmm_sync.db ue3.db inbox_notifications.db
do
    echo "========== $DB =========="
    sqlite3 -readonly "$BASE/$DB" ".schema"
done
========== gmm_myplaces.db ==========
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE sync_item (corpus INTEGER, key_string TEXT, timestamp BIGINT, merge_key BIGINT, feature_fprint BIGINT, latitude INTEGER, long
itude INTEGER, is_local BOOLEAN, sync_item BLOB, PRIMARY KEY (corpus, key_string));
CREATE INDEX idx_sync_item_timestamp ON sync_item (corpus, timestamp);
CREATE INDEX idx_sync_item_merge_key ON sync_item (corpus, merge_key);
CREATE INDEX idx_sync_item_feature_fprint ON sync_item (corpus, feature_fprint);
CREATE INDEX idx_sync_item_lat_long ON sync_item (corpus, latitude, longitude);
CREATE TABLE sync_corpus (corpus INTEGER PRIMARY KEY, last_sync_time BIGINT);
========== gmm_storage.db ==========
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE gmm_storage_table (_key_pri TEXT, _key_sec TEXT,_data BLOB,PRIMARY KEY(_key_pri, _key_sec));
========== ugc_photos_location_data.db ==========
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE photos_image_labels (photo_uri TEXT, image_label_mid INTEGER, image_label_confidence REAL NOT NULL, PRIMARY KEY (photo_uri,im
age_label_mid));
CREATE TABLE photos_top_feature (photo_uri TEXT PRIMARY KEY, latitude REAL, longitude REAL, valid_photo_taken_notification_photo BOOLEAN N
OT NULL, was_shown_in_photo_taken_notification BOOLEAN NOT NULL, was_uploaded BOOLEAN NOT NULL, is_face_detected BOOLEAN NOT NULL, timesta
mp LONG NOT NULL, was_dismissed_in_todo_list BOOLEAN NOT NULL);
========== gmm_sync.db ==========
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE sync_item_data (corpus INTEGER NOT NULL, client_id TEXT NOT NULL, server_id TEXT, timestamp BIGINT DEFAULT 0, feature_fprint
BIGINT, latitude_e6 INTEGER, longitude_e6 INTEGER, numerical_index BIGINT, string_index TEXT, sync_state INTEGER DEFAULT 0, item_proto BLO
B, PRIMARY KEY (corpus, client_id));
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -readonly -header -column "$BASE/ugc_photos_location_data.db" "
SELECT
    photo_uri,
    latitude,
    longitude,
    timestamp,
    datetime(timestamp/1000, 'unixepoch') AS date_time_utc,
    was_uploaded,
    is_face_detected
FROM photos_top_feature;
"

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -readonly "$BASE/ugc_photos_location_data.db" "
SELECT COUNT(*) AS total_photo_records
FROM photos_top_feature;
"
┌────────────────────┐
│ total_photo_records │
├────────────────────┤
│ 0                  │
└────────────────────┘
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -readonly -header -column "$BASE/gmm_sync.db" "
SELECT COUNT(*) AS total_records FROM sync_item_data;
"
total_records
─────────────
            7

┌──(kali㉿kali)-[~/pixel3]
└─$ sqlite3 -readonly -header -column "$BASE/gmm_myplaces.db" "
SELECT
    corpus,
    key_string,
    timestamp,
    datetime(timestamp/1000, 'unixepoch') AS date_time_utc,
    latitude,
    longitude,
    is_local
FROM sync_item;
"
corpus  key_string    timestamp       date_time_utc     latitude  longitude  is_local
──────  ────────────  ──────────────  ────────────────  ────────  ─────────  ────────
     8  1:0           1554407343604   2019-04-04 19:49:03  35734602  -78636654          0
```

```
    numerical_index,
    string_index,
    sync_state
FROM sync_item_data;
"
corpus    client_id       server_id     timestamp     date_time_utc     latitude_e6 longitude_e6 numerical_index  string_index    sync_state
──────    ──────────      ──────────    ──────────    ──────────────    ─────────── ──────────── ──────────────── ──────────────  ──────────
    11 4:0             4:0                       0 1970-01-01                35659596    -78872848                                           0
                                                    00:00:00

    11 5:0             5:0                       0 1970-01-01                40784374    -74065536                                           0
                                                    00:00:00

    11 1:0             1:0             1554407343604 2019-04-04                35734602    -78636654                                           0
                                                    19:49:03

    13 1581273852322_1                           0 1970-01-01                       0            0                1581273852322_1          0
       75453746935554                               00:00:00                                                     75453746935554

    13 1581273852322_1                           0 1970-01-01                       0            0                1581273852322_1          0
       75453747018211                               00:00:00                                                     75453747018211

    13 1581273852322_1                           0 1970-01-01                       0            0                1581273852322_1          0
       75453747040450                               00:00:00                                                     75453747040450

     6 User Parameters User Parameters           0 1970-01-01                       0            0                                          0
                                                    00:00:00
```

"

```
┌──(kali㉿kali)-[~/pixel3]
└─$ python3 - <<'PY'
locations = [
    ("Sync Record 1", 35659596, -78872848),
    ("Sync Record 2", 40784374, -74065536),
    ("Sync Record 3", 35734602, -78636654),
    ("Saved Place", 35734602, -78636654)
]

for name, lat, lon in locations:
    print(f"{name}: {lat/1000000}, {lon/1000000}")
PY
Sync Record 1: 35.659596, -78.872848
Sync Record 2: 40.784374, -74.065536
Sync Record 3: 35.734602, -78.636654
Saved Place: 35.734602, -78.636654
```

Investigate the image metadata

```
┌──(kali㉿kali)-[~/pixel3]
└─$ which exiftool
/usr/bin/exiftool

┌──(kali㉿kali)-[~/pixel3]
└─$ exiftool -r -csv \
-FileName \
-Directory \
-DateTimeOriginal \
-CreateDate \
-GPSLatitude \
-GPSLongitude \
"$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures" \
> "$HOME/pixel3/photo_metadata.csv"
    7 directories scanned
   11 image files read
```

Examine the actual image files

```
SourceFile                                                                                  FileName                  >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Skype/processed-1.jpeg                  processed-1.jpeg          >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/signal-2020-02-01-151617.jpg            signal-2020-02-01-151617.jpg>
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/LINE/1580586830715.jpg                  1580586830715.jpg         >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Telegram/IMG_20200203_210227_390.jpg    IMG_20200203_210227_390.jpg >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter/IMG_20200203_213026.jpg         IMG_20200203_213026.jpg    >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter/IMG_20200203_213034.jpg         IMG_20200203_213034.jpg    >
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213252.png  Screenshot_20200213-213252.p>
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213310.png  Screenshot_20200213-213310.p>
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200201-143429.png  Screenshot_20200201-143429.p>
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213151.png  Screenshot_20200213-213151.p>
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Messenger/received_517951378846562.jpeg    received_517951378846562.jpe>
(END)
```

Photographs with GPS information

```
┌──(kali㉿kali)-[~/pixel3]
└─$ exiftool -r -if '$GPSLatitude and $GPSLongitude' \
-FileName -GPSLatitude -GPSLongitude \
"$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures"
    7 directories scanned
   11 files failed condition
    0 image files read
```

Recovered images

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find "$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures" \
-type f \( -iname "*.jpg" -o -iname "*.jpeg" -o -iname "*.png" -o -iname "*.heic" \) \
! -path "*/.thumbnails/*" | wc -l
11
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find "$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures" \
-type f \( -iname "*.jpg" -o -iname "*.jpeg" -o -iname "*.png" -o -iname "*.heic" \) \
! -path "*/.thumbnails/*"
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Skype/processed-1.jpeg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/signal-2020-02-01-151617.jpg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/LINE/1580586830715.jpg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Telegram/IMG_20200203_210227_390.jpg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter/IMG_20200203_213026.jpg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter/IMG_20200203_213034.jpg
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213252.png
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213310.png
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200201-143429.png
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screenshots/Screenshot_20200213-213151.png
/home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Messenger/received_517951378846562.jpeg
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ find "$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures" \
-type f \( -iname "*.jpg" -o -iname "*.jpeg" -o -iname "*.png" -o -iname "*.heic" \) \
! -path "*/.thumbnails/*" \
-exec sha256sum {} \; \
> "$HOME/pixel3/photo_hashes.txt"

┌──(kali㉿kali)-[~/pixel3]
└─$ cat "$HOME/pixel3/photo_hashes.txt"
17dcc9092a09e98bb8ad7b24ec9f3948772bd307af9334eb76e8a58189e02a46  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Skype/p
rocessed-1.jpeg
f66260fd7c3188148730d51f3514df87c2a8a9e1cae4c3d8b1a428649880a16d  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/signal-
2020-02-01-151617.jpg
4de53429928191006ab29dbcec59a835b112cb0124179dafb1ea9707655867d4  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/LINE/15
80586830715.jpg
961b2194345c52ec1365310b822265da01a92db33cec89bfe93494d874635b98  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Telegra
m/IMG_20200203_210227_390.jpg
0f64608d14e0262be9a8c49edbe4982d0943dca0175fef2771d66f25895c25ee  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter
/IMG_20200203_213026.jpg
9e4cfa7156cee39e925ac25c29bc8a513b0b5264b4c18ebca3bba31ae5890491  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Twitter
/IMG_20200203_213034.jpg
50ed4b163f5a745877d5f2a736b84db9bcaef5247520b6a95e458156d4a0e1b8  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screens
hots/Screenshot_20200213-213252.png
bd6597d137b6f24d2fd84e1be8551272a2e08aa12373f28f6a20509487c67392  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screens
hots/Screenshot_20200213-213310.png
7d1dd8a56d360e18bf40296007d56e0330aa08dba6fdcf18ca49846fa13a84eb  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screens
hots/Screenshot_20200201-143429.png
ab89cddcabf4d95d7c4e3e585ced5da3b8796f8b7065f9af262ce01cc48f7130  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Screens
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ exiftool -csv -FileName -Directory -FileSize \
-FileModifyDate -DateTimeOriginal -CreateDate \
-GPSLatitude -GPSLongitude \
"$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Skype/processed-1.jpeg" \
"$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/signal-2020-02-01-151617.jpg" \
"$HOME/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/LINE/1580586830715.jpg" \
> "$HOME/pixel3/selected_photo_evidence.csv"
    3 image files read
```

```
┌──(kali㉿kali)-[~/pixel3]
└─$ grep -E 'processed-1.jpeg|signal-2020-02-01-151617.jpg|1580586830715.jpg' \
"$HOME/pixel3/photo_hashes.txt"
17dcc9092a09e98bb8ad7b24ec9f3948772bd307af9334eb76e8a58189e02a46  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/Skype/p
rocessed-1.jpeg
f66260fd7c3188148730d51f3514df87c2a8a9e1cae4c3d8b1a428649880a16d  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/signal-
2020-02-01-151617.jpg
4de53429928191006ab29dbcec59a835b112cb0124179dafb1ea9707655867d4  /home/kali/pixel3/pixel3_extracted/Pixel 3/data/media/0/Pictures/LINE/15
80586830715.jpg
```

9.1 Maps location entries: source and record type

Two Google Maps databases were examined:

Source database	Record type	Findings
gmm_myplaces.db	Saved place record	One location record found at latitude 35.734602, longitude -78.636654. Timestamp: 2019-04-04 19:49:03 UTC.
gmm_sync.db	Maps synchronisation records	Seven records found. Coordinates included 35.659596, -78.872848, 40.784374, -74.065536, and 35.734602, -78.636654.

Two of the coordinates in gmm_sync.db had zero timestamps. One coordinate matched the saved place record in gmm_myplaces.db.

9.2 Photographs: paths, hashes, capture times, offset and GPS

Eleven image files were identified in the extracted media folder. The following three were selected for metadata examination:

Photograph and path	SHA-256	File modification time	Capture time	Offset	GPS
Pictures/Skype/processed-1.jpeg	17dcc9092a09e98bb8ad7b24ec9f3948772bd307af9334eb76e8a58189e02a46	2020-02-01 19:47:46	Not available	-05:00 on file modification time	None
Pictures/Signal/signal-2020-02-01-151617.jpg	f66260fd7c3188148730d51f3514df87c2a8a9e1cae4c3d8b1a428649880a16d	2020-02-01 15:16:30	Not available	-05:00 on file modification time	None
Pictures/LINE/1580586830715.jpg	4de53429928191006ab29dbcec59a835b112cb0124179dafb1ea9707655867d4	2020-02-01 14:53:50	Not available	-05:00 on file modification time	None

The three files had no DateTimeOriginal, CreateDate, or GPS metadata. The recorded times are file modification times, not confirmed photograph capture times.

9.3 Maps-to-photo comparison attempted

The Google Maps location records were compared with the available photograph metadata.

- The Maps databases contained location coordinates.
- The examined photographs did not contain GPS coordinates.
- The photographs did not contain reliable capture-time metadata.
- The saved Maps coordinate at 35.734602, -78.636654 could not be linked to any of the examined photographs.

9.4 Strength and limitation of the comparison

Strength: The examination recovered location records from Google Maps and identified image files with their SHA-256 hashes and file modification times.

Limitation: The absence of GPS and original capture-time metadata prevented a reliable location and time match between the Maps records and photographs.

A saved place or location record, together with photograph metadata, does not prove that a particular person visited that location. The records only show data stored on the examined device.

Conclusion

Google Maps contained saved location and synchronisation records. Eleven image files were identified, and the examined photographs lacked GPS and original capture-time metadata. A direct link between the Maps locations and the photographs could not be established from the available evidence.

10. Task 8 - Timeline of Events

Raw Time	Field/Unit	UTC Conversion	Event Description	Evidence ID
2019-04-04 19:49:03	Maps saved-place date/time	2019-04-04 19:49:03 UTC	A saved place was recorded in Google Maps at coordinates 35.734602, -78.636654.	T7-01
2020-02-01 14:53:50 (-05:00)	File modification time	2020-02-01 19:53:50 UTC*	LINE image file modification time.	T7-02
2020-02-01 15:16:30 (-05:00)	File modification time	2020-02-01 20:16:30 UTC*	Signal image file modification time.	T7-03
2020-02-01 19:40:13	Kik message timestamp	2020-02-01 19:40:13 UTC	Incoming Kik message from Josh Hickman.	T6-01
2020-02-01 19:42:36	Kik message timestamp	2020-02-01 19:42:36 UTC	Outgoing Kik reply about Kik remaining online.	T6-01
2020-02-01 19:43:27	Kik message timestamp	2020-02-01 19:43:27 UTC	Incoming Kik reply: "Great (sarcasm)."	T6-01
2020-02-01 19:46:25	Kik message timestamp	2020-02-01 19:46:25 UTC	Outgoing Kik message discussing video calling and ads.	T6-01
2020-02-01 19:47:04	Kik message timestamp	2020-02-01 19:47:04 UTC	Incoming Kik message with blank message body.	T6-01
2020-02-01 19:47:46 (-05:00)	File modification time	2020-02-02 00:47:46 UTC*	Skype image file modification time.	T7-04
2020-02-08 20:55:14	WhatsApp message timestamp	2020-02-08 20:55:14 UTC	Outgoing WhatsApp message: "Hi there! I switched over."	T6-02
2020-02-08 20:56:05	WhatsApp message timestamp	2020-02-08 20:56:05 UTC	Incoming WhatsApp reply: "Awesome!"	T6-02
2020-02-08	WhatsApp	2020-02-08	Outgoing WhatsApp	T6-02

Raw Time	Field/Unit	UTC Conversion	Event Description	Evidence ID
20:57:49	message timestamp	20:57:49 UTC	message: "The Osbourne!"	
2020-02-09 18:41:42	SMS timestamp	2020-02-09 18:41:42 UTC	Outgoing SMS to 9195790479 about going to a coffee shop.	T4-01
2020-02-14 13:09:03	SMS timestamp	2020-02-14 13:09:03 UTC	Outgoing SMS about messages syncing between accounts.	T4-01
1581644358241	User creation time (milliseconds, Unix epoch)	2020-02-14 01:39:18.241 UTC	Android User 11 account creation time.	T2-01
1581684601922	Last login time (milliseconds, Unix epoch)	2020-02-14 12:50:01.922 UTC	Last recorded login time for Android User 11.	T2-01
1581695517124	Last login time (milliseconds, Unix epoch)	2020-02-14 15:51:57.124 UTC	Last recorded login time for Android User 0.	T2-01

The photo conversions use the recorded -05:00 offset. These are file modification times, not confirmed photo capture times.

B. Conclusions

The examination of the Pixel 3 extraction revealed activity across SMS, Google Messages, WhatsApp, Kik, Google Maps, Android user accounts and image files.

The timeline shows communication activity on 1 February, 8 February, 9 February and 14 February 2020. Kik messages were exchanged with a contact identified as Josh Hickman on 1 February. WhatsApp messages were also exchanged with the same contact name in the contact records on 8 February. SMS records show further communication on 9 and 14 February.

The examination also found two Android user profiles, saved Google Maps location data and image files from Skype, Signal and LINE. The examined photographs did not contain GPS coordinates or confirmed camera capture times.

Some records share contact details or names, but the available evidence does not prove that every account or communication was operated by the same person. A saved Maps location does not prove that the device owner physically visited that location.

C. Overall Finding

The extracted data provides evidence of communication and device activity during the examined period. The findings support a timeline of recorded events, but they do not independently establish the identity of the device operator or prove the physical location of the user at a given time.

11. Conclusion and Limitations

Category	Content
Direct observations (what the records literally show)	The extraction contains Android 10 system data identifying the device as a Google Pixel 3. Two Android user profiles, User 0 and User 11, were recorded.
Direct observations (communication records)	SMS, MMS, call records and messaging application databases contain communication activity. WhatsApp records show messages dated 8 February 2020. Kik records show six message entries dated 1 February 2020. SMS records include outgoing messages dated 9 and 14 February 2020.
Direct observations (contacts and accounts)	The name Josh Hickman and related contact details appear in Android contacts and messaging application records. The WhatsApp contact database also contains a JID linked to the phone number recorded for Josh Hickman.
Direct observations (Maps and photographs)	Google Maps databases contain saved-place and synchronisation records with location coordinates. Eleven image files were found in the examined Pictures folders. The three selected photographs had file modification times but no GPS tags or confirmed camera capture times.
Reasonable inferences (what you conclude, and why)	The device was used for communication through several applications during the period covered by the recovered records. The dates and message contents support a sequence of recorded communication activity.
Reasonable inferences (contact relationship)	The repeated appearance of Josh Hickman's name and related contact details across contacts, WhatsApp, Kik and call records supports a possible relationship between the records. However, it does not independently prove the identity of the person using each account.
Alternative interpretations (at least two)	**1. Shared device:** A shared company or personal device may have been used by more than one person at different times. Therefore, account and profile activity does not confirm the conduct of one individual. **2. Synchronised or imported data:** Some contact, message or Maps records may have been synchronised, restored or imported from another device or account. Their presence does not always prove that the activity took place directly on the examined phone. **3. Saved location versus physical visit:** Google Maps coordinates may represent saved or synchronised places rather than locations

	physically visited by the device user.
Additional evidence that would resolve the question	Further evidence could include the original device for a physical examination, account login and synchronisation records, Google account activity records, cloud backups, application server records, network connection logs, and reliable location records with timestamps. Additional evidence linking a person to the device at the relevant times would also help establish who performed the recorded activities.

11.2 Evidence IDs

Evidence ID	Evidence source
T2-01	Android user accounts and system settings
T4-01	SMS database (mmssms.db)
T6-01	Kik messages database
T6-02	WhatsApp messages database
T7-01	Google Maps saved-place database
T7-02	LINE image metadata
T7-03	Signal image metadata
T7-04	Skype image metadata

11.3 Overall Conclusion

The examination recovered useful digital evidence from the Google Pixel 3 extraction. The records show activity involving multiple communication applications, Android user profiles, contact records, Google Maps data and image files.

The recovered timestamps support a timeline of stored events. However, some records may have been synchronised or restored, and some timestamps represent file modification times rather than confirmed event or capture times.

The available evidence is not enough to prove that one named individual performed every recorded action or physically visited the saved Maps locations.

The findings should therefore be treated as evidence of activity recorded in the extracted data, subject to the limitations stated in this report.

12. Executive Summary

This examination was conducted on an Android 10 extraction from a Google Pixel 3 device under case reference **SBT-DF205-C1**. The purpose was to examine the available device data, identify user and device information, recover communication records, review application activity, inspect Google Maps data and photographs, and build a timeline of relevant events.

The examination covered Android system information, user profiles, device settings, SMS, MMS, contacts, call logs, calendar records, Google Messages, Google Dialer, WhatsApp, Kik, Google Maps databases and selected image files.

The findings confirmed two Android user profiles, identified communication records across several messaging applications, and recovered messages exchanged with a contact saved as Josh Hickman. The examination also found saved Google Maps location data and image files from Skype, Signal and LINE.

The examined photographs did not contain GPS coordinates or confirmed camera capture times. The Maps records showed saved or synchronised location data, but did not prove that the device or its user physically visited those locations.

The findings provide a record of activity stored in the extracted data. However, they do not independently identify the person who operated the device at every point in time or prove that one person was responsible for all recorded activity.